



GPGPU - Workshop (General Purpose GPU)

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22.06.2026



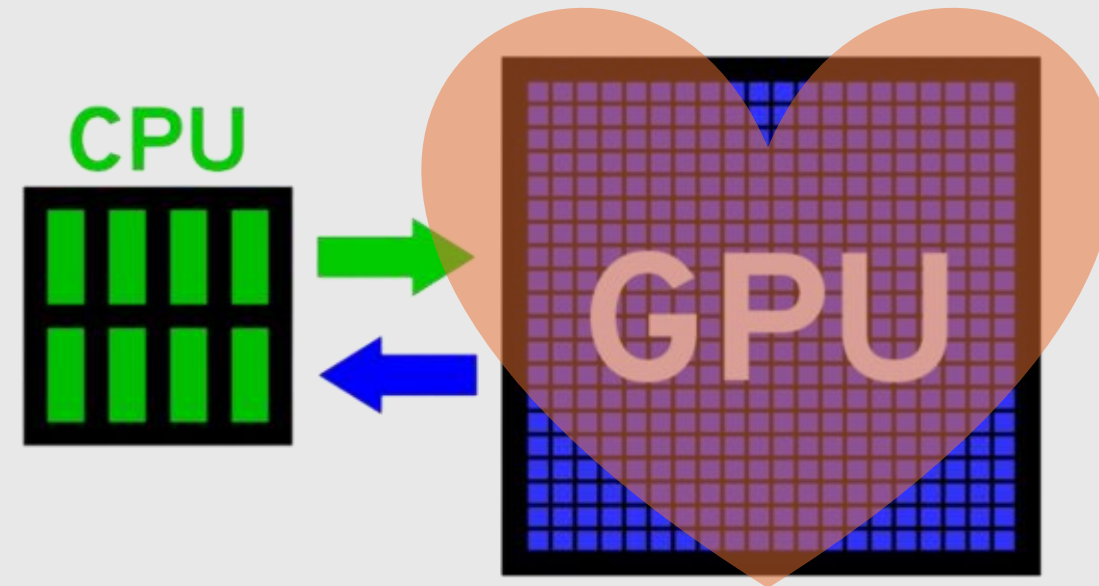
GPGPU – Ce reprezintă ?

- Metoda **GPGPU**

- **G**eneral **P**urpose **G**raphics **P**rocessing **U**nit
- Utilizarea procesorului grafic pentru operațiuni generale

- **Rezultate**

- Paralelism incredibil
- Dezvoltare de biblioteci care ne fac viața mai bună



GPGPU – Realizare metodă

- Implementare metodă
 - API CUDA
 - Programăm prima oară GPU-ul în scopuri generale



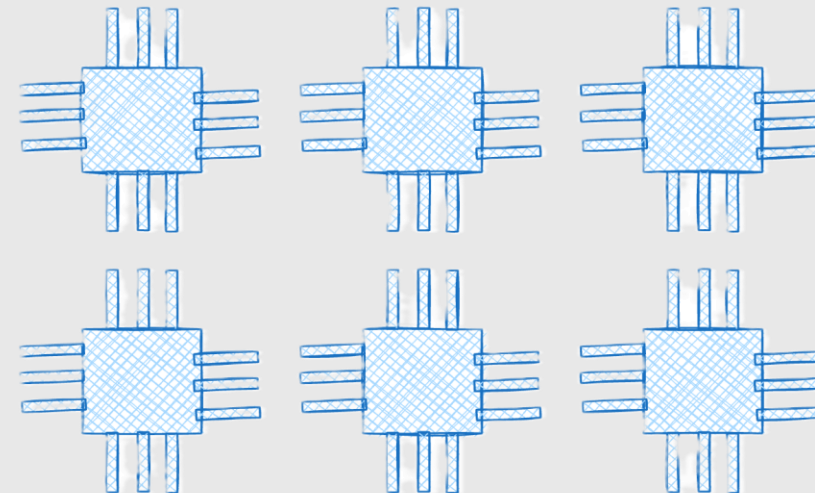
NVIDIA CUDA



CUDA



GPU



USE-CASES



Speech

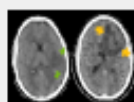


Translate



Recommender

CONSUMER INTERNET



Healthcare



Manufacturing

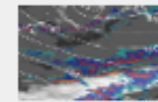


Finance

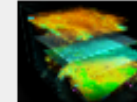
INDUSTRIAL APPLICATIONS



Molecular
Simulations



Weather
Forecasting



Seismic
Mapping

SUPERCOMPUTING

APPS & FRAMEWORKS



Amber

NAMD

+600
Applications

CATIA



AUTODESK
3DS MAX

Ps



Windows 10

CUDA-X LIBRARIES

MACHINE LEARNING

cuDF

cuML

cuGRAPH

DL / HPC

cuDNN

CUTLASS

TENSORRT

CUDA Math Libraries

LANGUAGES

python OpenACC



LLVM Compiler
For CUDA

CUDA

CUDA TOOLKIT

CUDA
COMPILER

DEVELOPER TOOLS

DEBUGGERS

PROFILERS

CUDA C++
CORE

CUDA DRIVER

MEMORY
MANAGEMENT

WINDOWS &
GRAPHICS

COMMS
LIBRARIES

OS PLATFORMS



CentOS

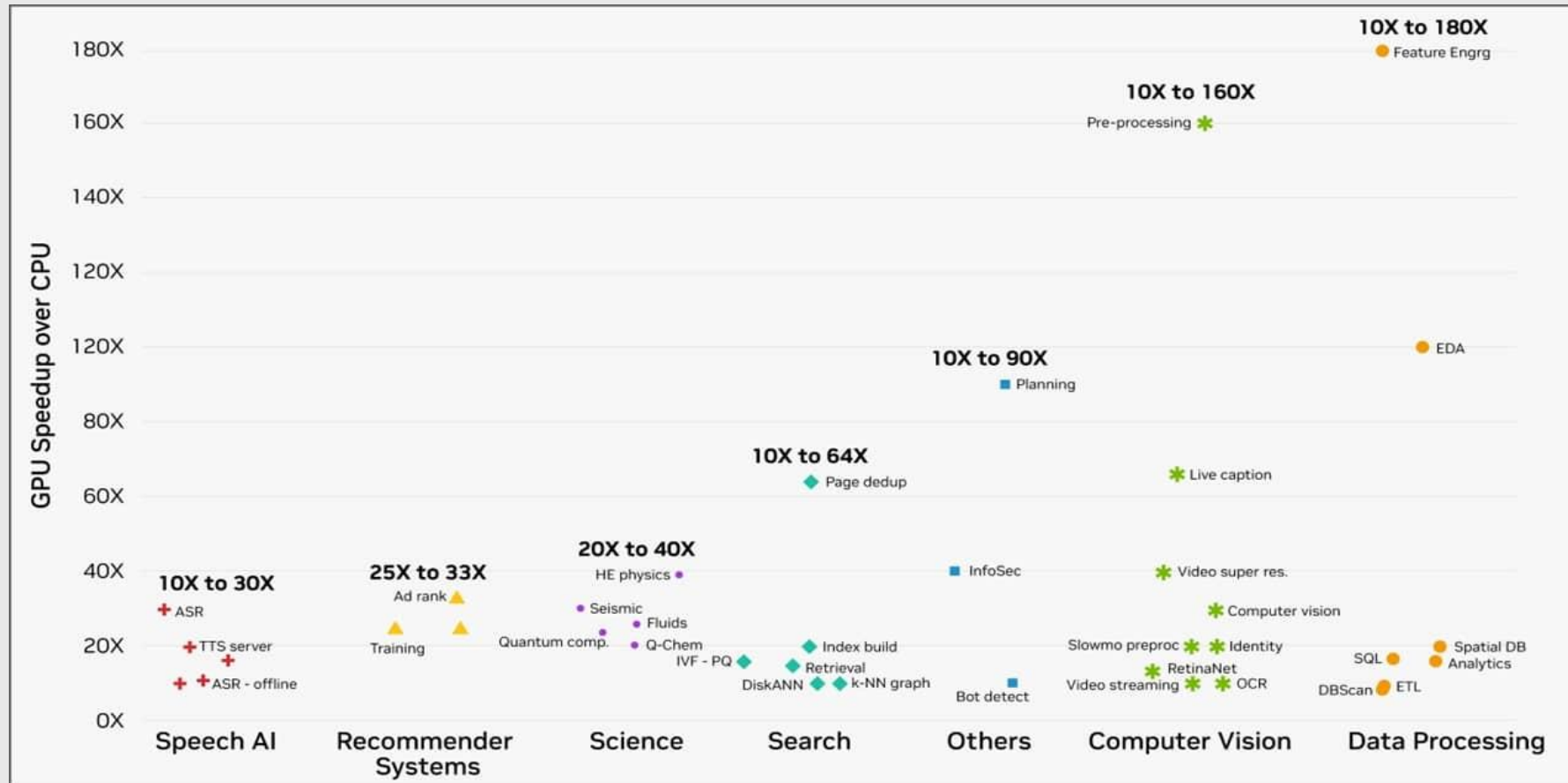


Windows Server

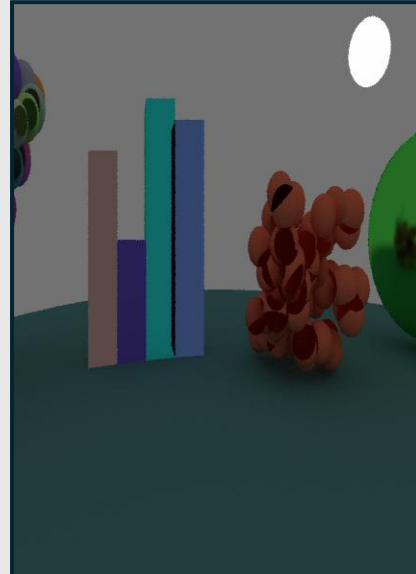
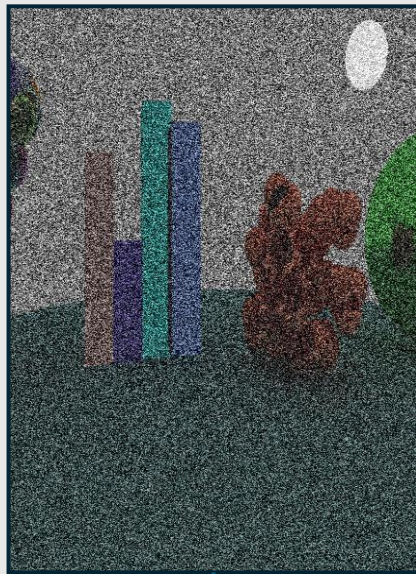


Accelerare

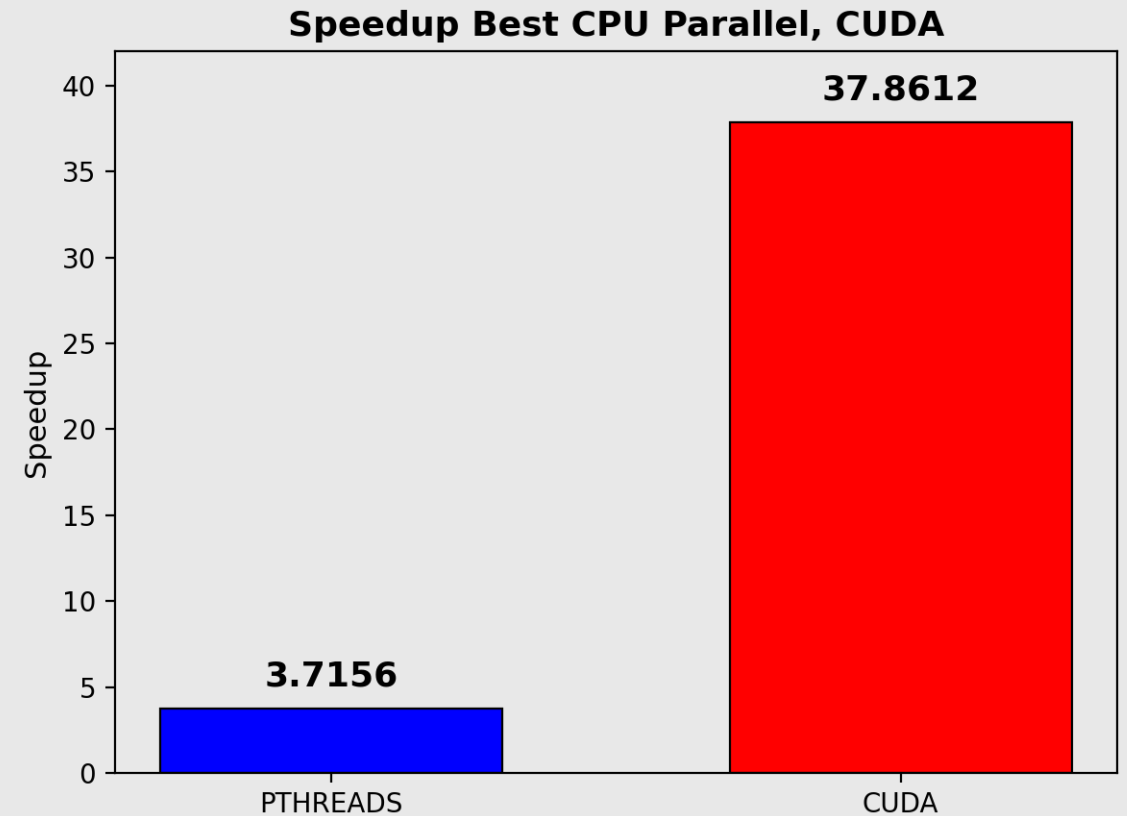
*“Make it Work. Make it Right. **Make it Fast**”(Kent Beck)*



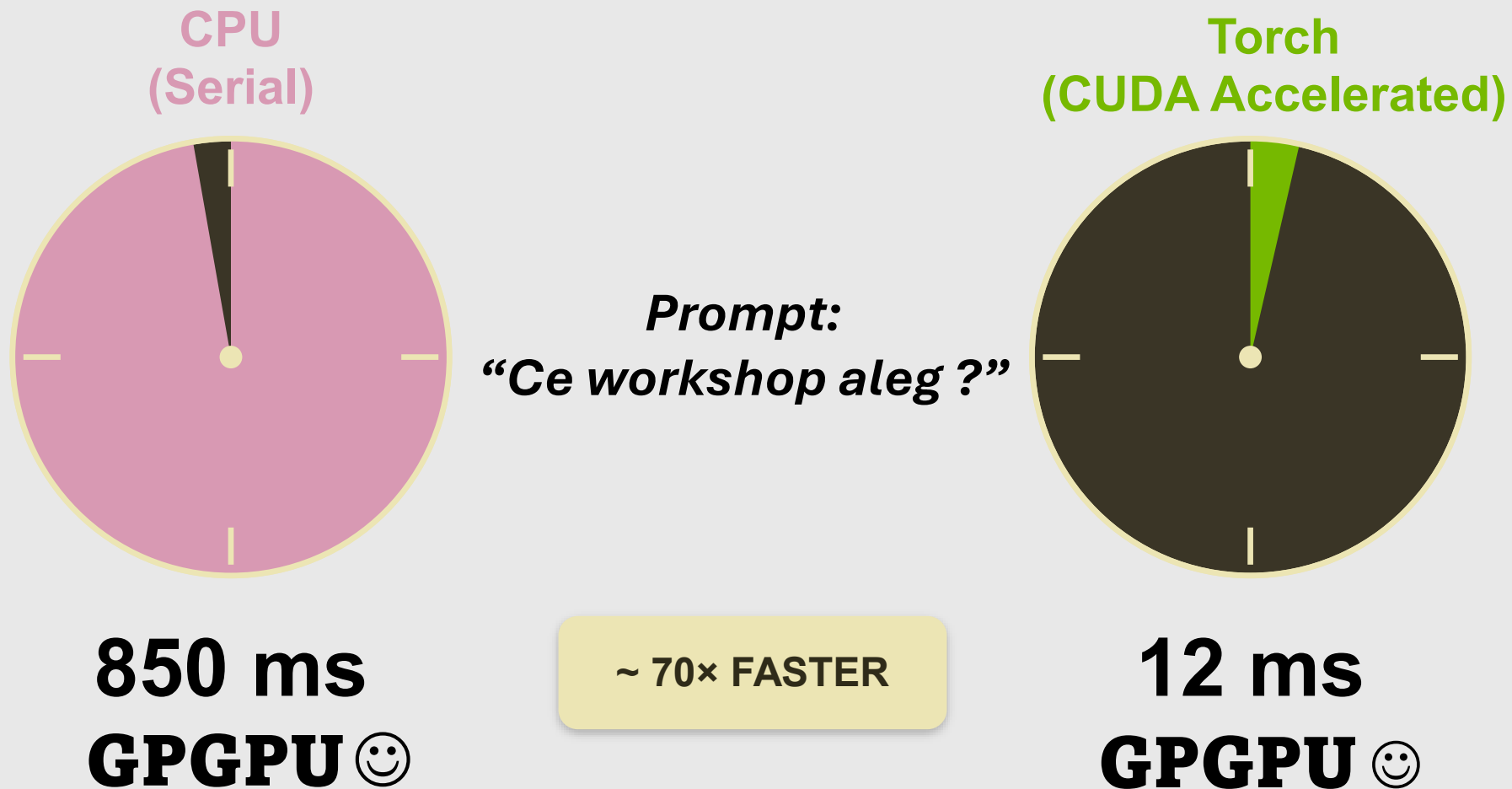
Exemplu practic – Procesare de imagini



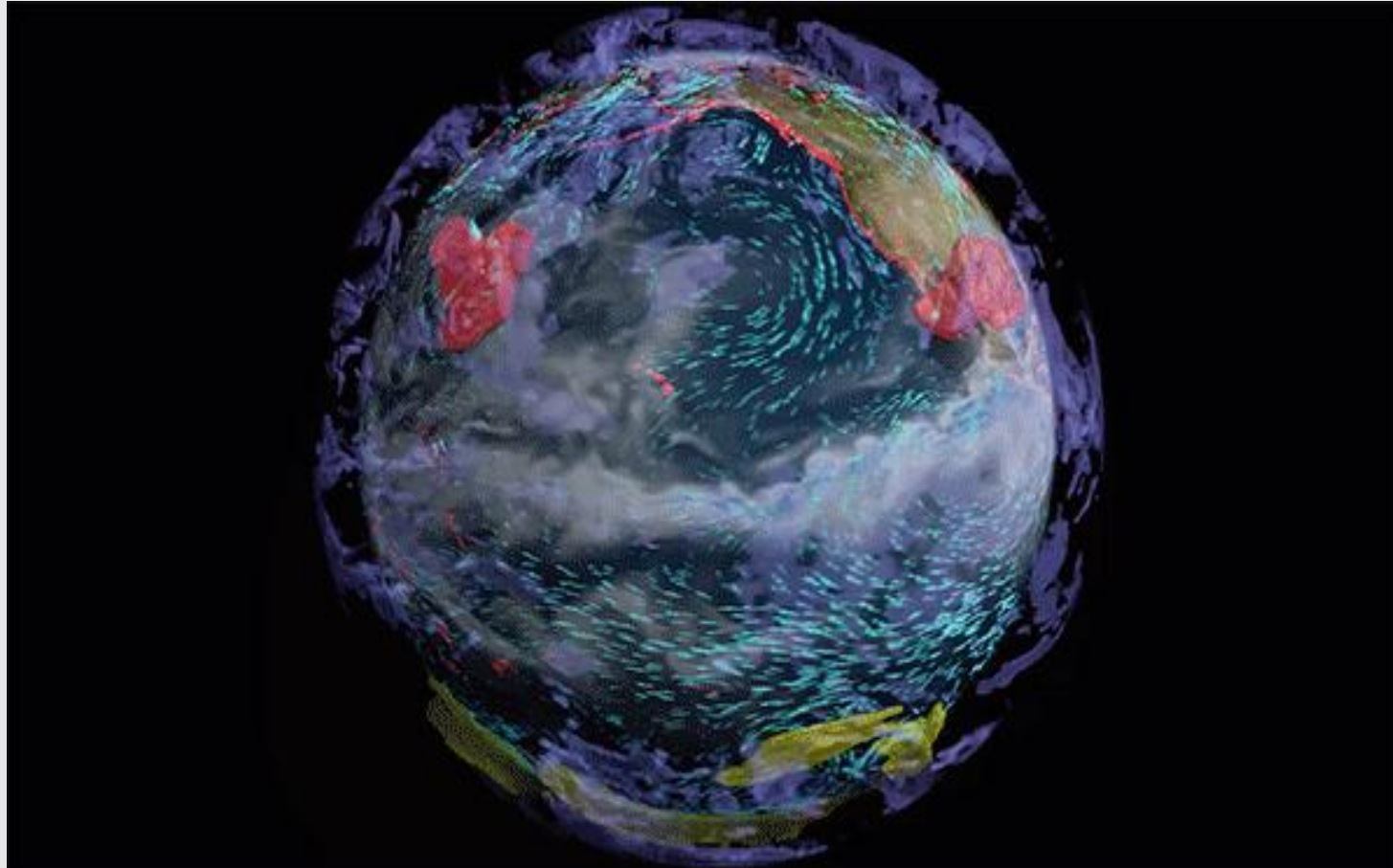
Eliminare zgomot



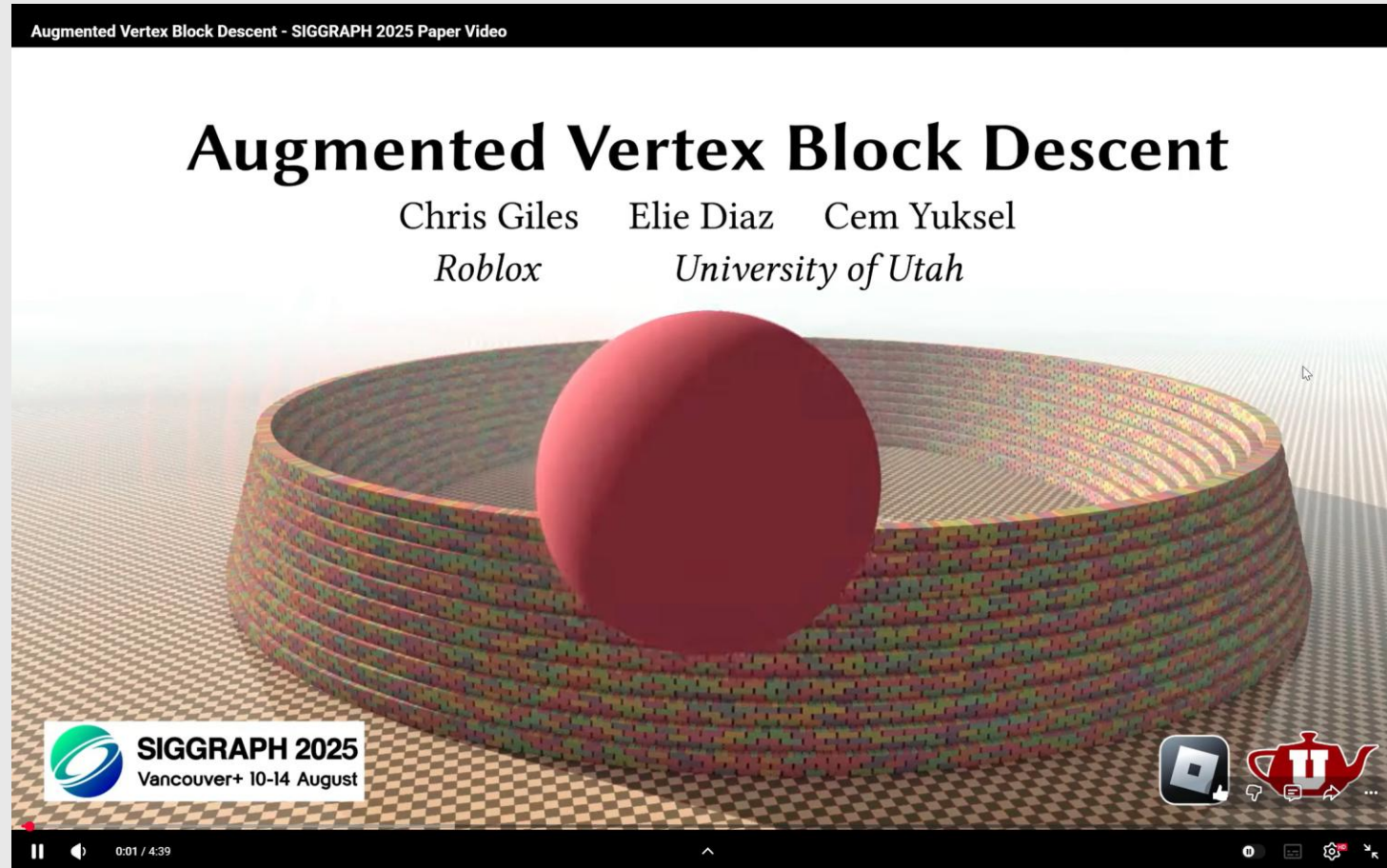
Exemplu practic – Inferență model I.A.



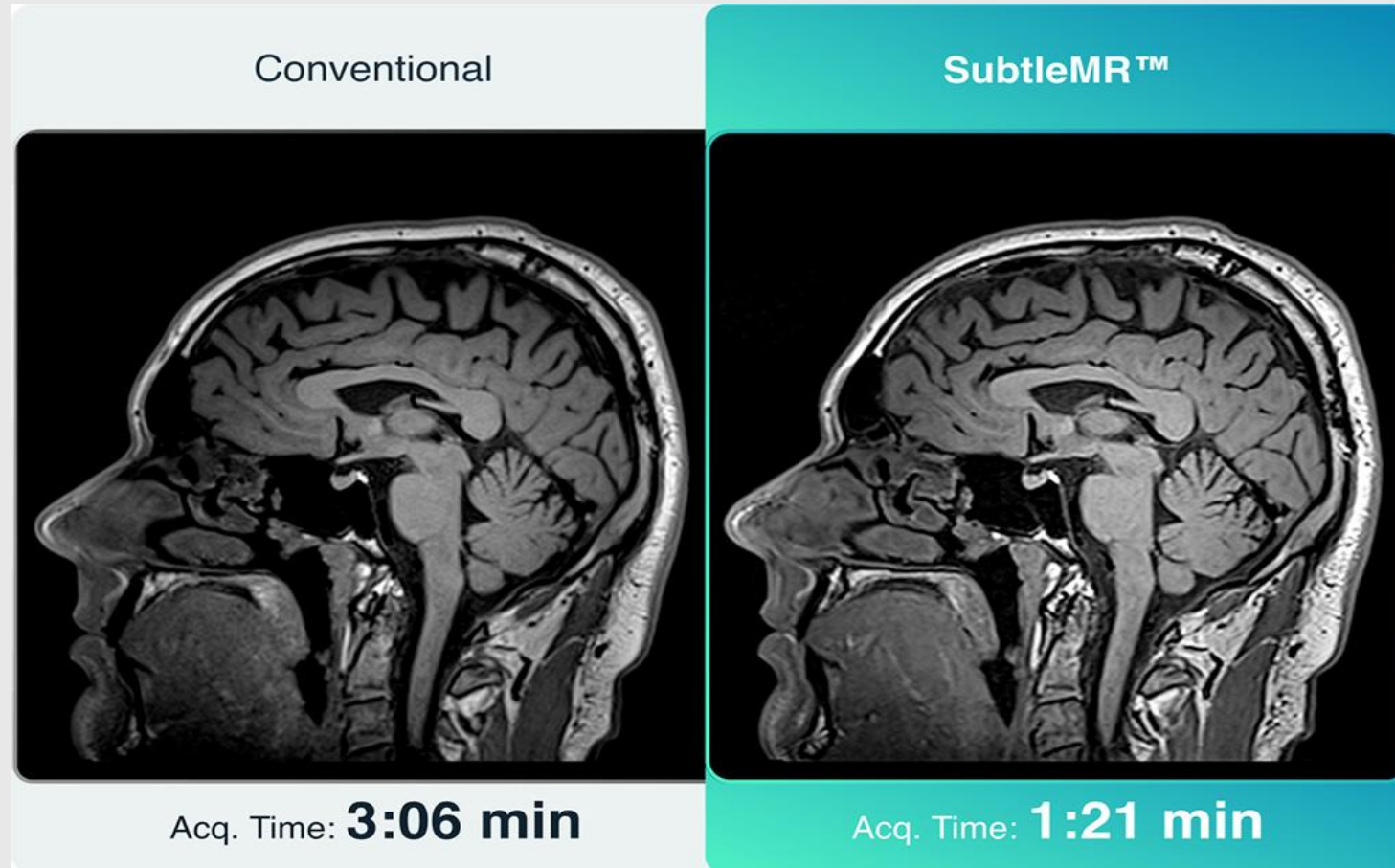
...și mai multe utilități - Meteorologie



...și mai multe utilități – Simulări fizice



...și mai multe utilități - Medicină



Rezumat Workshop

- **Program**

- 4 întâlniri începând cu 20 iulie (2 ore fiecare)
- Date stabilite împreună
- Introducere în arhitectura GPU și metoda GPGPU
- Exerciții practice CUDA
- Proiect

- **Mediu de lucru**

- GPU NVIDIA: lucru pe sistemul propriu
- Altele: lucru pe cluster-ul facultății

- **Rezultate**

- Înțelegerea arhitecturii GPU
- Înțelegerea programării paralele pe un mediu masiv paralel

Mulțumesc!
Vă aștept la workshop!